

Haoyu Wang

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EDUCATION

University of Pittsburgh

Aug 2025 – 2029 (Expected)

Ph.D in Biomedical Informatics & Artificial Intelligence

- **Focus:** Computational Biology, Machine Learning, Multi-Omics

Beijing Forestry University

Sep 2020 – Jun 2024

Bachelor of Engineer in Computer Science and Technology

EXPERIENCE

University of Pittsburgh Medical Center

Aug 2024 – Jul 2025

Computational Biology Intern

Pittsburgh, PA

- Performed downstream pathway and transcription factor activity inference using statistical modeling and enrichment analysis to identify key regulatory signals in tumor microenvironments
- Designed and implemented dual-graph attention neural networks (DGAT) for spatial multi-omics data integration, enabling accurate protein expression imputation from transcriptomic and spatial features

PUBLICATIONS

Wang, H., Batz, C., Osmanbeyoglu, H.U., *DGAT: A Dual-Graph Attention Network for Inferring Spatial Protein Landscapes from Transcriptomics*. *Nat Commun* **17**, 6702, (2025). doi: <https://doi.org/10.1038/s41467-026-73114-z>
Abstract selected for oral presentation at ISMB 2025; workshop presentation at MLCB 2025; tutorial at ECCB 2026.

Linan Zhang, April Sagan, Bin Qin, **Haoyu Wang**, Elena Kim, Baoli Hu, Hatice Ulku Osmanbeyoglu, *STAN, a computational framework for inferring spatially informed transcription factor activity*, *Nucleic Acids Research*, Volume 54, Issue 1, 13 January 2026, gkaf1473. doi: [10.1093/nar/gkaf1473](https://doi.org/10.1093/nar/gkaf1473)

PROJECTS

DGAT: Spatial Protein Landscapes Imputation from Transcriptomics | Computational Framework Development 2025

- Designed and developed a **spatially-aware GAT-based model for protein imputation** from mRNA and spatial data.
- Improved protein-wise correlation by **5%–20% over 7 prior methods** in sample-hold-out benchmarking.
- In spatial transcriptomic data lacking protein level but with pathological annotations, achieved at least **1%–3% Adjusted Rand Index** gains in spatial domain clustering.

Spatially Fine-Tuned Geneformer for Spatial Transcriptomics Domain Adaptation | Foundation Model Adaptation 2026

- Adapted **Geneformer v2**, a single-cell foundation model, to spatial transcriptomics using **LoRA-based parameter-efficient fine-tuning** with spatial contrastive learning.
- Designed a spatially aware fine-tuning framework by constructing spatial KNN graphs and incorporating neighborhood-aware representation learning to improve tissue-level embedding quality.
- Improved spatial domain identification on human lymph node and TNBC datasets, achieving **F1 score improvement from 0.159 to 0.816** and **ARI improvement from 0.219 to 0.531** compared with original Geneformer embeddings.
- Demonstrated enhanced pathology-aware representation learning, where fine-tuned Geneformer embeddings achieved comparable classification performance to mRNA features while substantially outperforming pretrained embeddings.

STAN: a Computational Framework Inferring Transcription Factor Activity | Image Feature Extraction 2024

- Applied unsupervised deep learning methods to **whole slide image feature extraction** for downstream modeling.
- Identification and visualization of significant patterns to uncover biological insights related to cancer progression.

TECHNICAL SKILLS

Programming Languages: Proficient – Python, SQL; Intermediate – Java, C++, MATLAB, R; Basic

Machine Learning/AI: PyTorch, TensorFlow, scikit-learn, GNNs, Transformer, Computer Vision

Bioinformatics: Spatial Multi-Omics, Single-cell Analysis, Pathway Modeling, Pathology Image Analysis